

# FURTHER MECHANICS

## Mock Paper 2 (Set B)

Time Allowed: 1 hour 30 minutes • Total Marks: 75

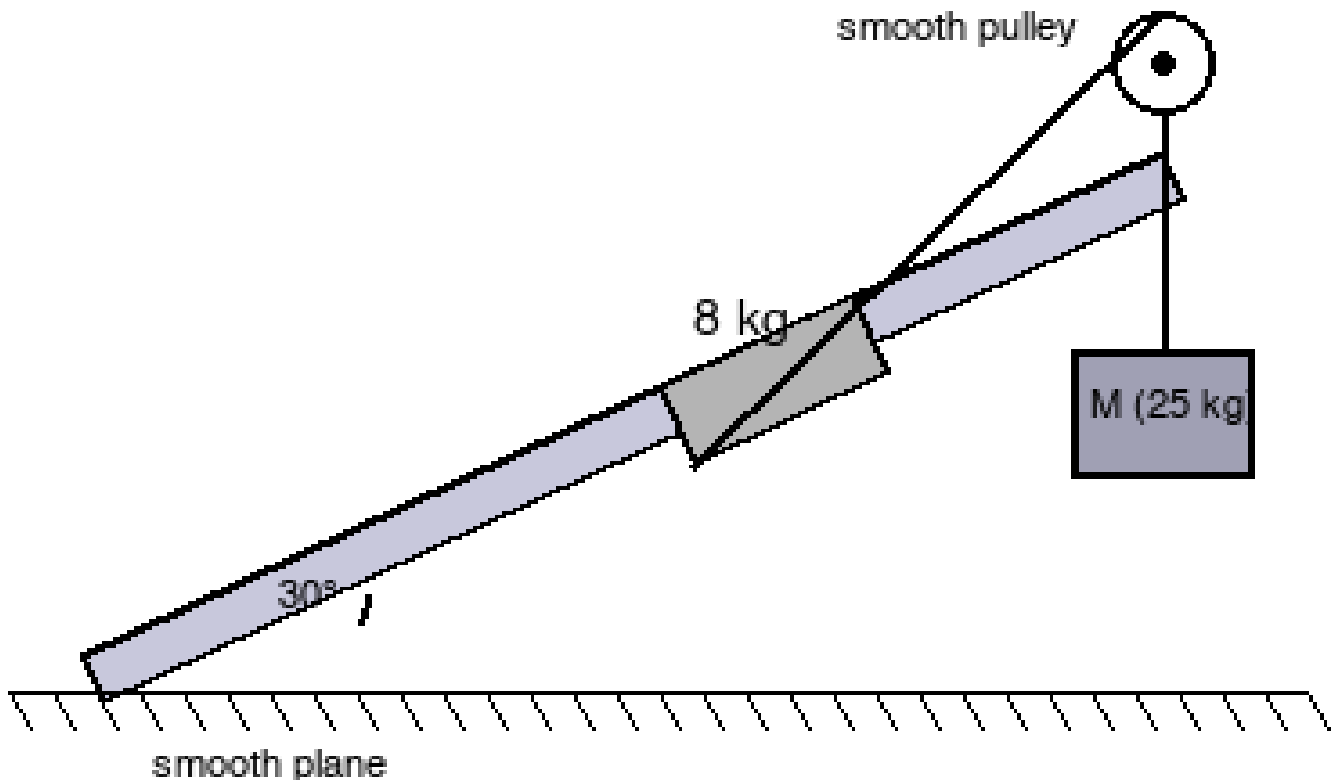
Candidate Name \_\_\_\_\_ Candidate Number \_\_\_\_\_

### Instructions

- Use black or dark blue pen. Pencil may be used for diagrams.
- Answer **all** questions.
- Unless otherwise stated, take  $g = 9.8 \text{ m s}^{-2}$  and give answers to an appropriate degree of accuracy.
- Non-exact numerical answers should be given to 3 significant figures unless stated otherwise.
- The total number of marks for this paper is 75.
- You are permitted to use a scientific calculator.

### 1. [11 marks]

A light inextensible string passes over a small smooth pulley fixed at the top of a smooth plane inclined at  $30^\circ$  to the horizontal. An 8 kg block lies on the plane and is attached to one end of the string. A particle of mass  $M$  kg hangs vertically below the pulley at the other end. The system is shown in the diagram below.

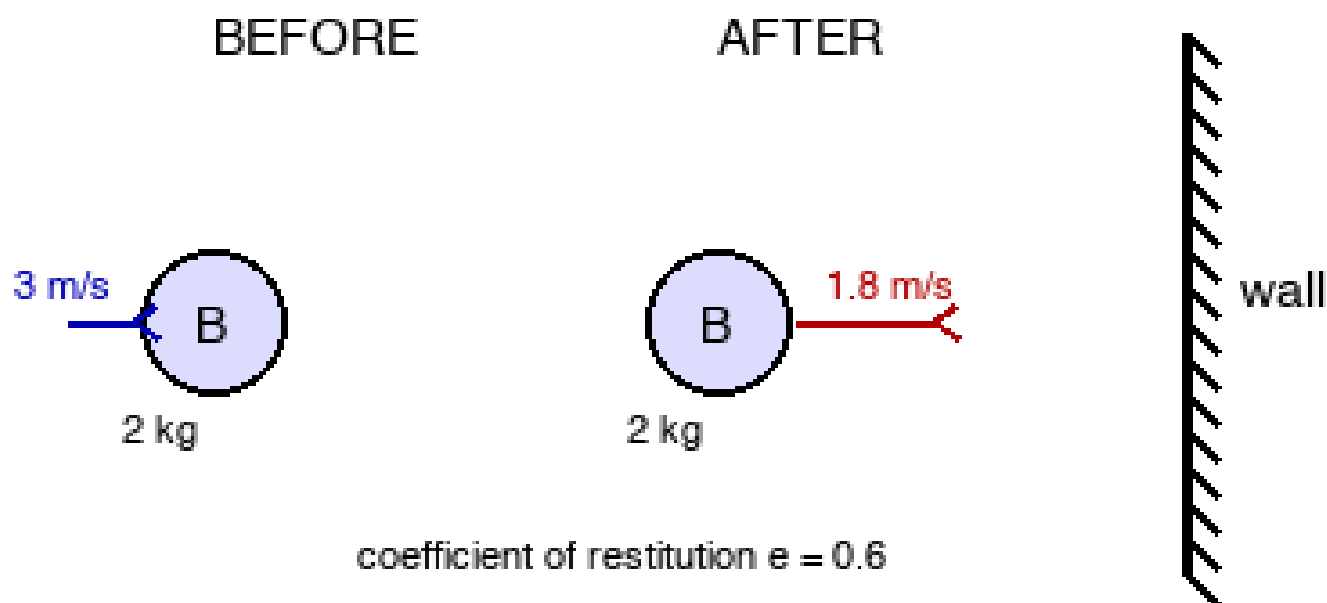


The system is released from rest with the string taut.

- Find, in terms of  $M$ , the acceleration of the 8 kg block up the plane. [3]
- Given that the acceleration of the block is  $2.1 \text{ m s}^{-2}$ , find the value of  $M$ . [3]
- Find the tension in the string. [2]
- The string is now cut. State, with a reason, whether the 8 kg block will begin to slide back down the plane. [3]

### 2. [8 marks]

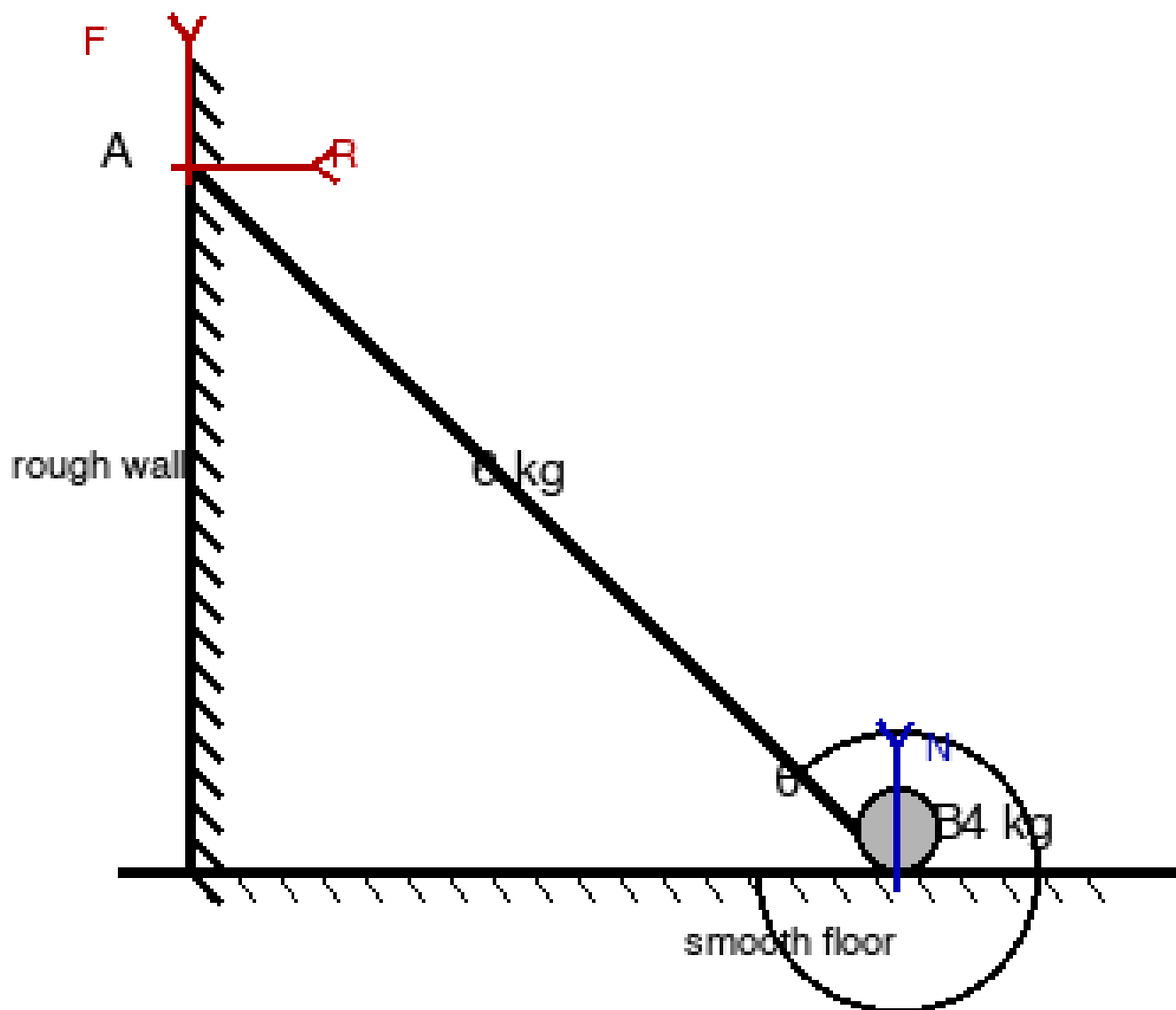
A particle A of mass 2 kg moving with speed  $3 \text{ m s}^{-1}$  directly towards a smooth vertical wall collides with the wall. The coefficient of restitution between the particle and the wall is 0.6. The situation is shown in the diagram below.



- Find the speed of the particle after the impact. [2]
- Find the impulse exerted by the wall on the particle, stating its direction. [3]
- A second particle B of mass 3 kg approaches the same wall with speed  $4 \text{ m s}^{-1}$  and rebounds with speed  $2 \text{ m s}^{-1}$ . Find the coefficient of restitution for this second collision. [3]

### 3. [12 marks]

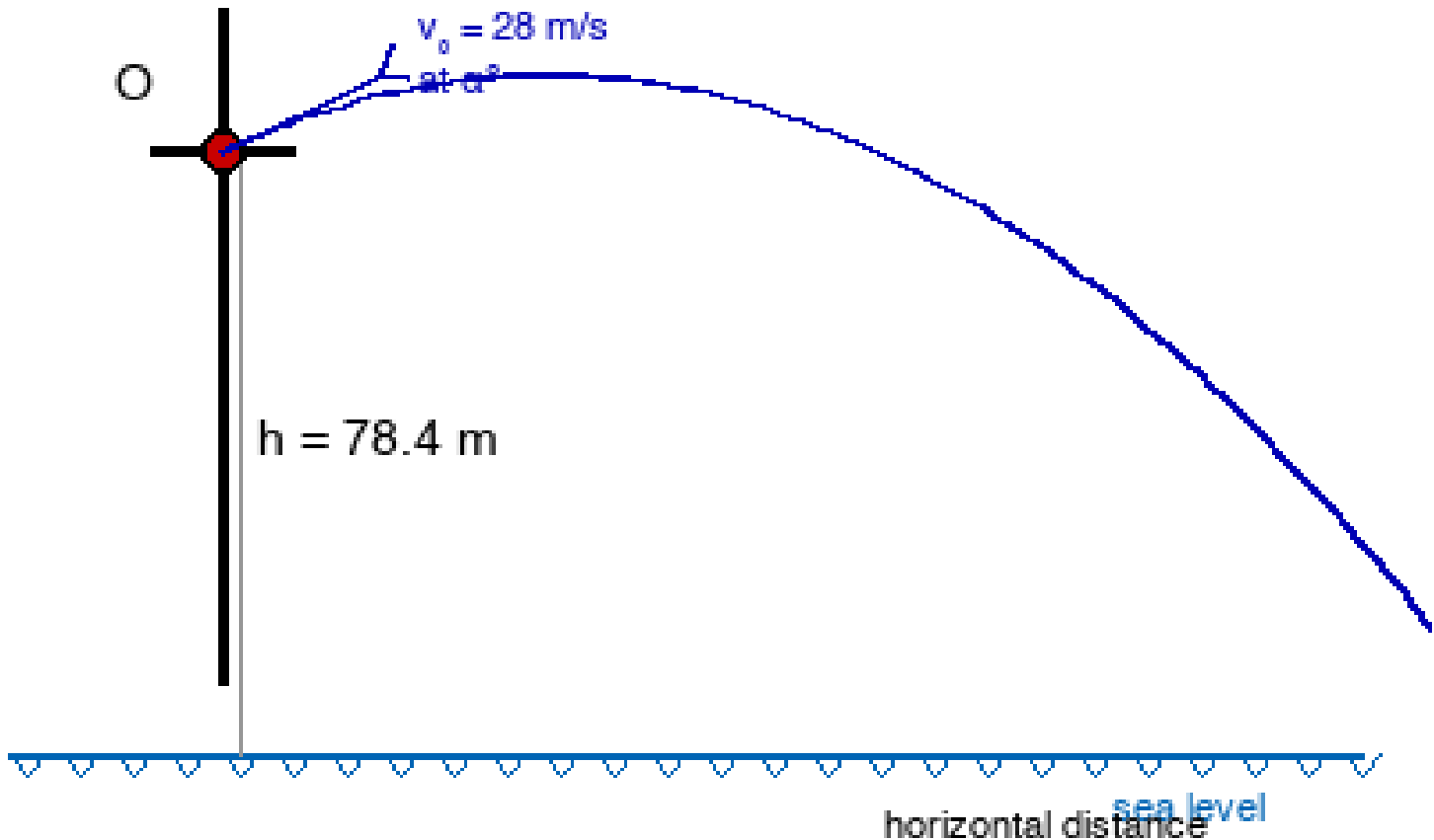
A uniform rod AB of mass 6 kg and length 5 m rests with end A against a rough vertical wall and end B on a smooth horizontal floor. The rod makes an angle  $\theta$  with the horizontal. A particle of mass 4 kg is attached to B. The system is shown in the diagram below.



- (a) By taking moments about  $A$ , show that the normal reaction at  $B$  is 52.1 N (to 3 s.f.). [4]  
 (b) Find the magnitude of the frictional force at  $A$ . [3]  
 (c) Given that the rod is on the point of slipping, find the coefficient of friction between the rod and the wall. [3]  
 (d) State, with a reason, whether the rod would begin to slip if the 4 kg particle were placed at the midpoint of  $AB$  instead of  $B$ . [2]

#### 4. [12 marks]

A particle is projected from a point  $O$  at the top of a vertical cliff of height  $78.4$  m above sea level, with speed  $28 \text{ m s}^{-1}$  at an angle  $\alpha$  above the horizontal. The particle hits the sea at point  $P$  as shown in the diagram below.



- (a) Given that  $\alpha = 20^\circ$ , find the time taken for the particle to reach the sea. [3]
- (b) Find the horizontal distance  $OP$ . [2]
- (c) Find the speed and direction of motion of the particle when it hits the sea at  $P$ . [4]
- (d) Find the value of  $\alpha$  for which the horizontal distance  $OP$  is a maximum. [3]

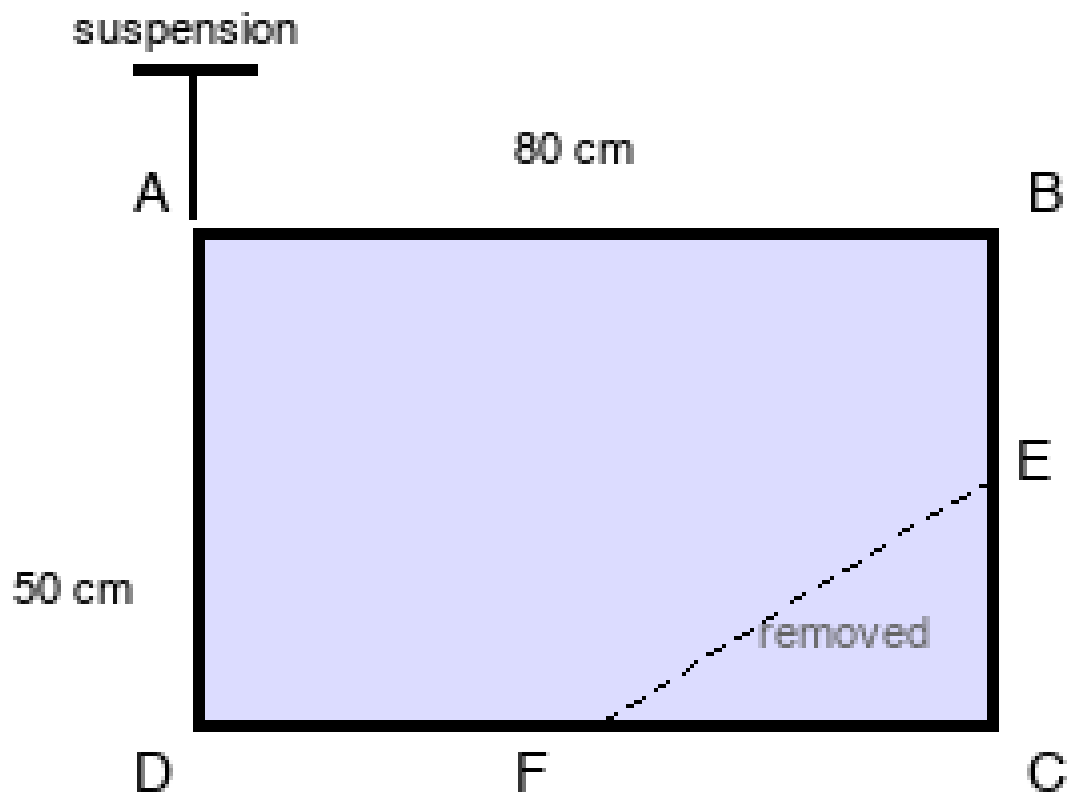
#### 5. [9 marks]

Two particles  $A$  and  $B$ , of masses  $4 \text{ kg}$  and  $6 \text{ kg}$  respectively, are connected by a light inextensible string passing over a small smooth pulley fixed at the top of a rough plane inclined at  $40^\circ$  to the horizontal. Particle  $A$  lies on the plane and particle  $B$  hangs vertically below the pulley. The coefficient of friction between  $A$  and the plane is  $0.3$ . The string lies along a line of greatest slope of the plane. The system is released from rest.

- (a) Find the acceleration of particle  $B$ . [4]
- (b) Find the tension in the string. [2]
- (c) When particle  $B$  has fallen a distance of  $2 \text{ m}$ , the string breaks. Find the total distance travelled by particle  $A$  from the moment of release until it first comes to rest. [3]

#### 6. [10 marks]

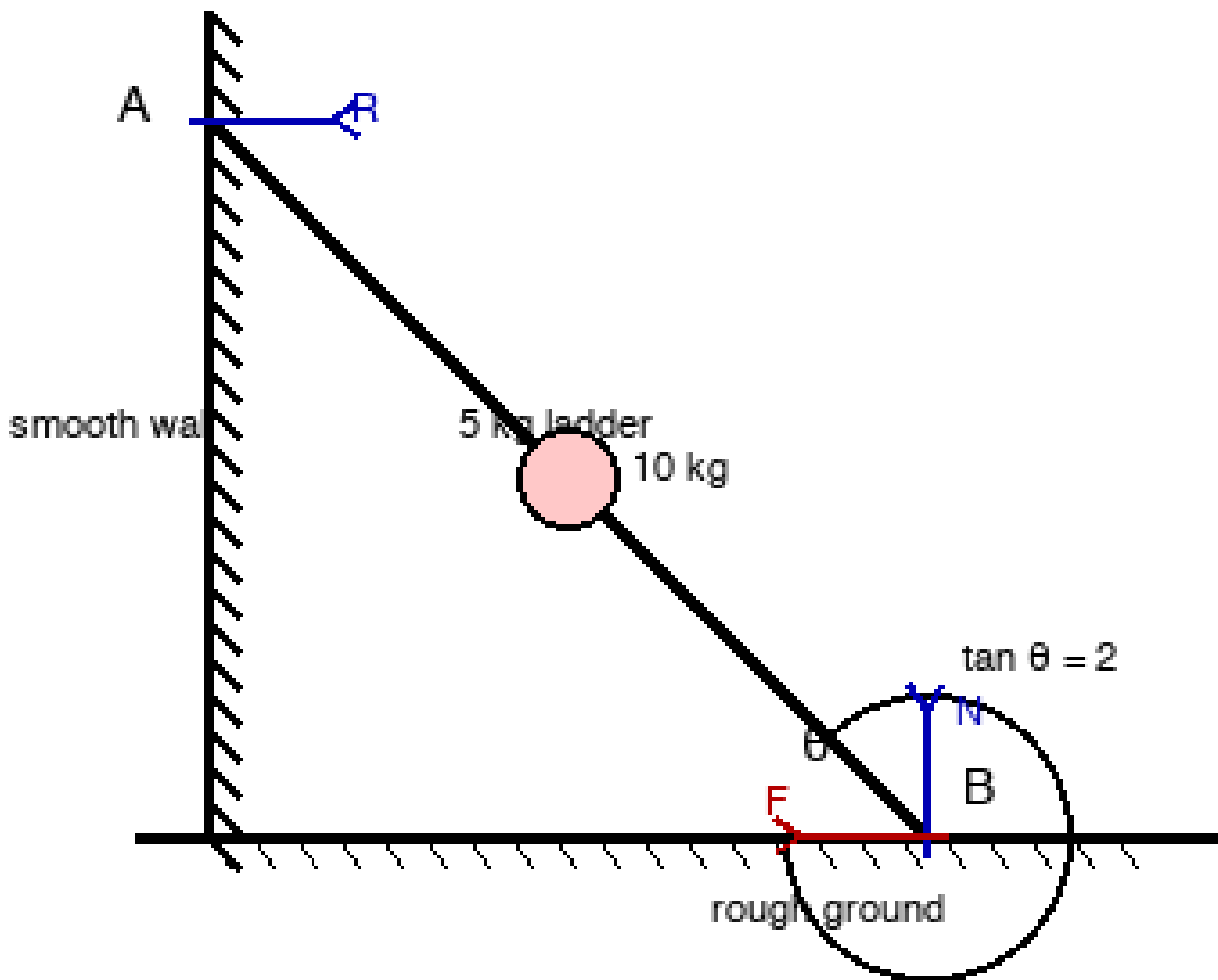
A uniform rectangular lamina  $ABCD$  has  $AB = 80 \text{ cm}$  and  $AD = 50 \text{ cm}$ . A small triangular portion  $CEF$  is removed, where  $E$  and  $F$  are the midpoints of  $BC$  and  $CD$  respectively. The lamina is suspended from  $A$ . The diagram below shows the lamina and the removed portion.



- Show that the centre of mass of the original rectangle  $ABCD$  is at the point  $(40, 25)$  from  $A$ . [1]
- Find the coordinates of the centre of mass of the triangular portion  $CEF$  from  $A$ . [2]
- Hence find the coordinates of the centre of mass of the remaining lamina from  $A$ . [4]
- Find the angle that  $AB$  makes with the vertical when the lamina hangs in equilibrium from  $A$ . [3]

#### 7. [13 marks]

A uniform ladder  $AB$  of mass  $5\text{ kg}$  and length  $4\text{ m}$  rests in limiting equilibrium with end  $A$  against a smooth vertical wall and end  $B$  on rough horizontal ground. A person of mass  $10\text{ kg}$  stands at the midpoint of the ladder. The ladder makes an angle  $\theta$  with the horizontal, where  $\tan \theta = 2$ . The set-up is shown in the diagram below.



- (a) By resolving horizontally and vertically, show that the normal reaction at  $A$  equals the frictional force at  $B$ , and that the normal reaction at  $B$  is 147 N. [3]
- (b) By taking moments about  $B$ , find the normal reaction at  $A$ . [4]
- (c) Hence find the coefficient of friction between the ladder and the ground. [3]
- (d) The person now climbs up the ladder towards  $A$ . Explain, without any further calculation, whether the ladder will begin to slip before the person reaches  $A$ . [3]

**END OF PAPER**